

Class 09: Candy Mini-Project

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Background

In this mini project we will take a dive into some exploratory techniques within **r**, including creating correlation plots and PCA plots to explore and intepret a dataset on candy.

```
candydata<- "candy-data.csv"
candy = read.csv(candydata, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294

One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
table(candy$fruity)
```

```
0 1
47 38
```

There are 38 different fruity types of candy.

What is your favorite candy?

The percentage of people who prefer a certain candy over a randomly chosen one is called \$winpercent.

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy["Milky Way", ]$winpercent
```

```
[1] 73.09956
```

The winpercent value for Milky Way is 73%

Q4. What is the winpercent value for "Kit Kat"?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

The winpercent for Kit Kat is around 77%

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

The winpercent for Tootsie Roll Snack Bars is around 50%

Now we will dive into the `skim()` function that is apart of the **skimr** package. This helps to summarize and overview the dataset.

```
skimr::skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The **winpercent** column is on an entirely different scale than the other columns, while the other columns fall between 0-1, the **winpercent** column falls between 0-100.

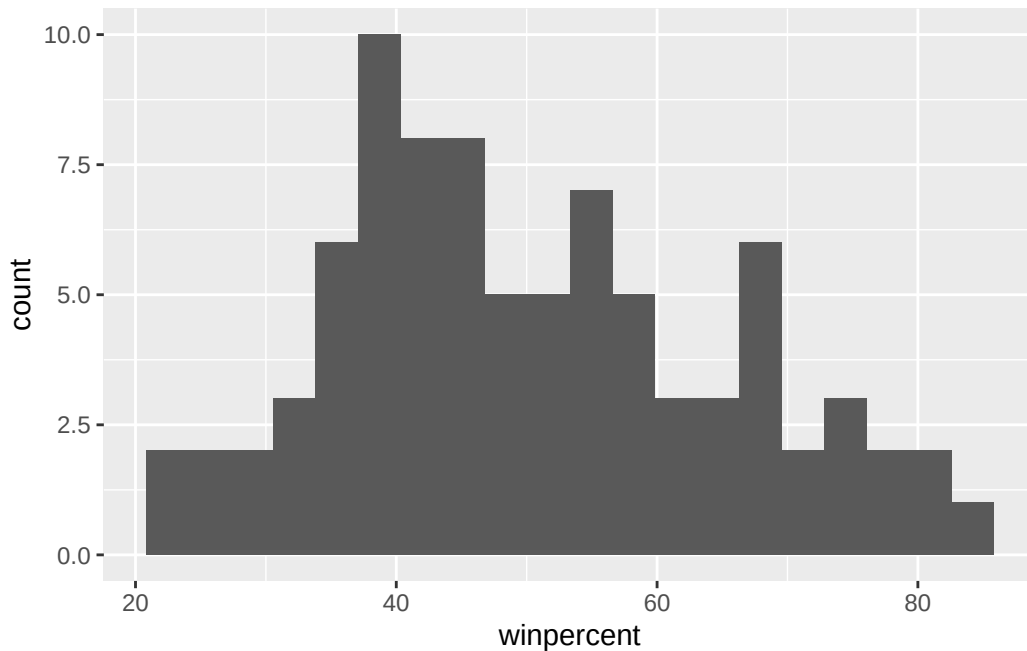
Q7. What do you think a zero and one represent for the `candy$chocolate` column?

The presence or absence of chocolate within that particular candy. A 1 designating that candy has chocolate and a 0 designating that it is chocolate free.

Exploratory Analysis

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
library(ggplot2)
ggplot(candy)+
  aes(winpercent)+
  geom_histogram(bins=20)
```



Q9. Is the distribution of winpercent values symmetrical?

No they are not symmetrical, it appears that they are skewed to the right.

Q10. Is the center of the distribution above or below 50%?

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

The mean is slightly above 50, while the median is lower than 50. Median is a better tell of the true center of the data.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
choc.win<- candy[as.logical(candy$chocolate==1),"winpercent"]
fruit.win<- candy[as.logical(candy$fruity==1),"winpercent"]
mean(choc.win)
```

```
[1] 60.92153
```

```
mean(fruit.win)
```

```
[1] 44.11974
```

```
mean(choc.win)>mean(fruit.win)
```

```
[1] TRUE
```

On average chocolate candy is ranked higher than fruit candy.

Q12. Is this difference statistically significant?

```
t.test(choc.win, fruit.win)
```

Welch Two Sample t-test

```
data:  choc.win and fruit.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

p-value = 2.871e-08, so they are significantly different because they are below the alpha of 0.05.

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
# Here we are using dplyr
library(dplyr)
candy |>
  arrange(winpercent)|>
  select(winpercent)|>
  head(5)
```

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

The least liked candies are Nik L Nip, Boston backed beans, Chiclets, supper bubble and jaw busters.

```
#Here we are using base r
head(candy[order(candy$winpercent),])
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0
Root Beer Barrels	0	0	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197	0.976	
Boston Baked Beans				0	0	0	1	0.313	0.511	
Chiclets				0	0	0	1	0.046	0.325	
Super Bubble				0	0	0	0	0.162	0.116	
Jawbusters				0	1	0	1	0.093	0.511	
Root Beer Barrels				0	1	0	1	0.732	0.069	

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744
Root Beer Barrels	29.70369

Q14. What are the top 5 all time favorite candy types out of this set?

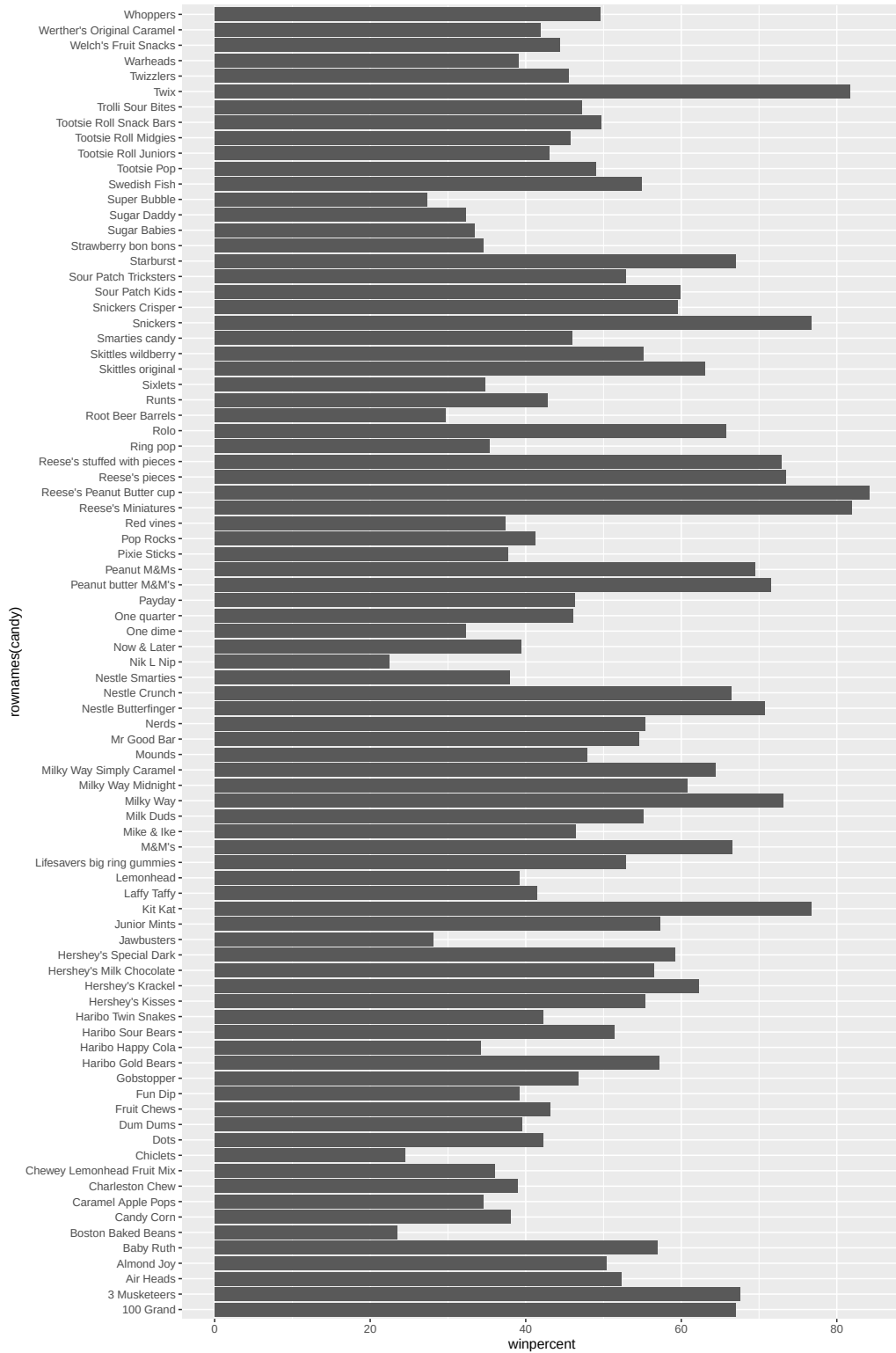
```
candy |>
  arrange(-winpercent)|>
  select(winpercent)|>
  head(5)
```

	winpercent
Reese's Peanut Butter cup	84.18029
Reese's Miniatures	81.86626
Twix	81.64291
Kit Kat	76.76860
Snickers	76.67378

The top 5 are Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, Snickers

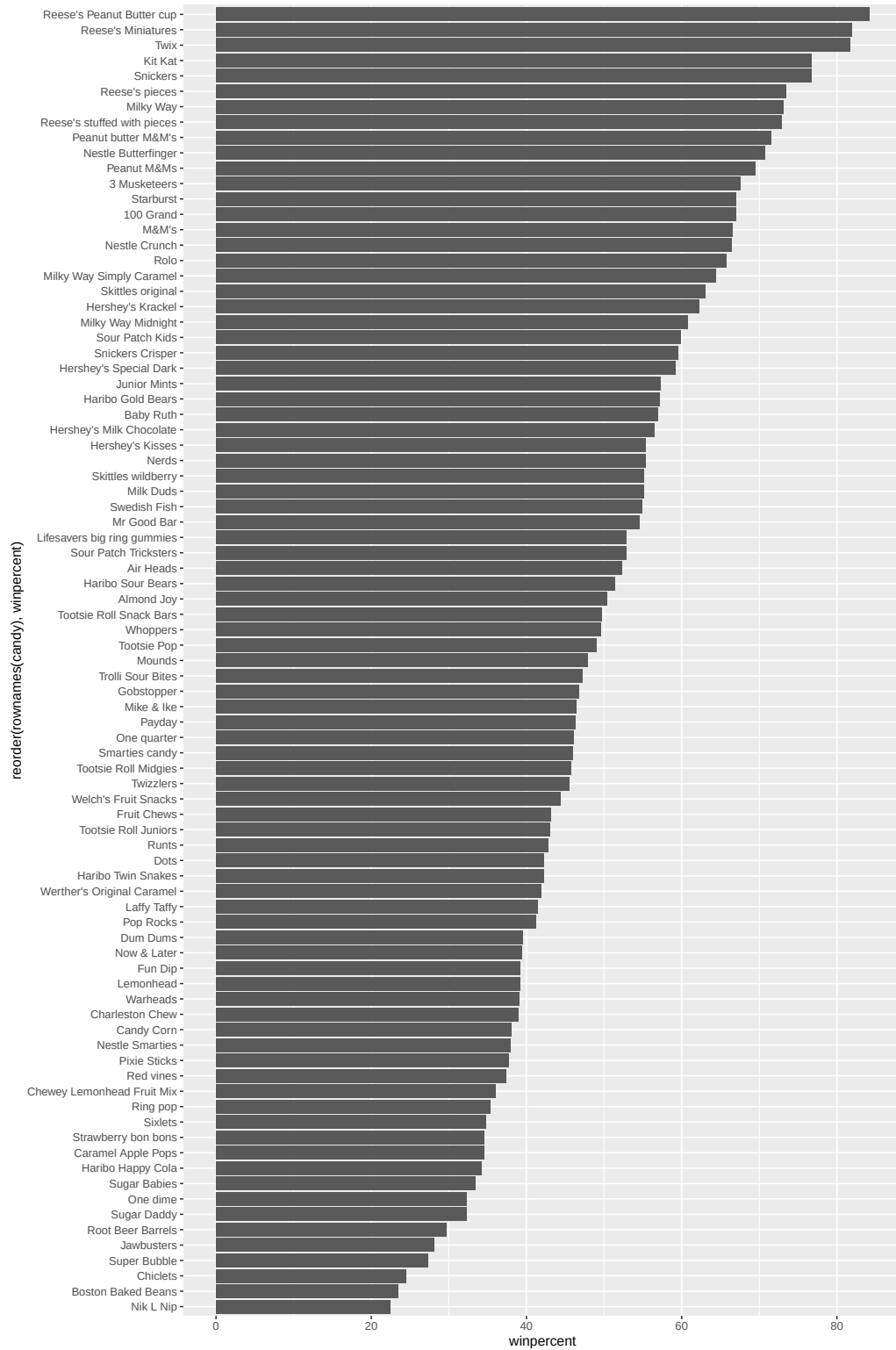
Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy)+  
  aes(winpercent, rownames(candy))+  
  geom_col()
```

Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent? Tip

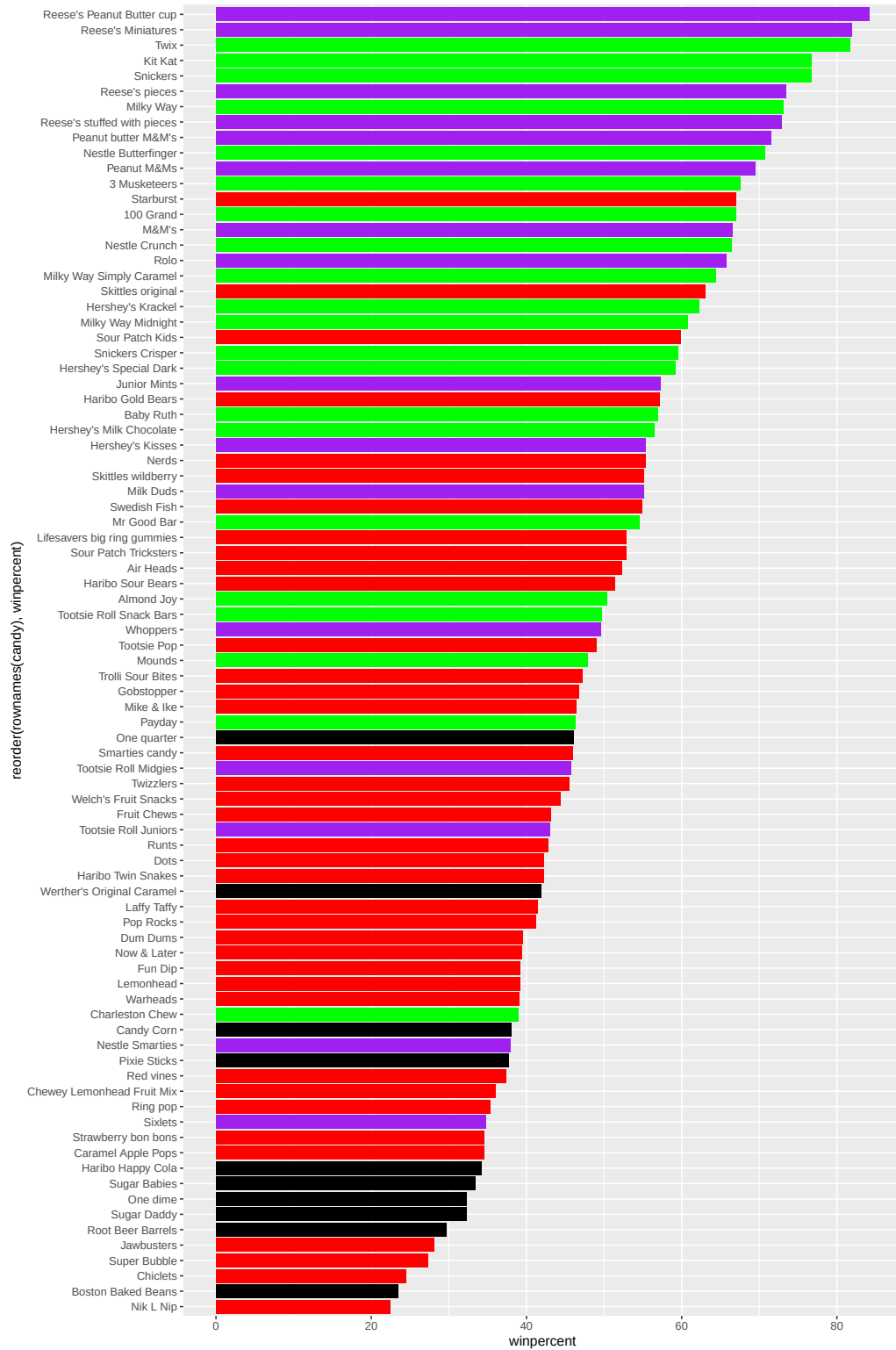
```
ggplot(candy)+  
  aes(winpercent, reorder(rownames(candy), winpercent))+  
  geom_col()
```



Adding Color

```
my_cols=rep("black",nrow(candy))
#Chocolate= Purple
my_cols[as.logical(candy$chocolate)]<- "purple"
#Bar= Yellow
my_cols[as.logical(candy$bar)]<- "green"
#Fruity= Red
my_cols[as.logical(candy$fruity)]<- "red"
```

```
ggplot(candy)+
  aes(winpercent, reorder(rownames(candy), winpercent))+
  geom_col(fill=my_cols)
```



Q17. What is the worst ranked chocolate candy?

The Worst ranked chocolate candy is sixlets

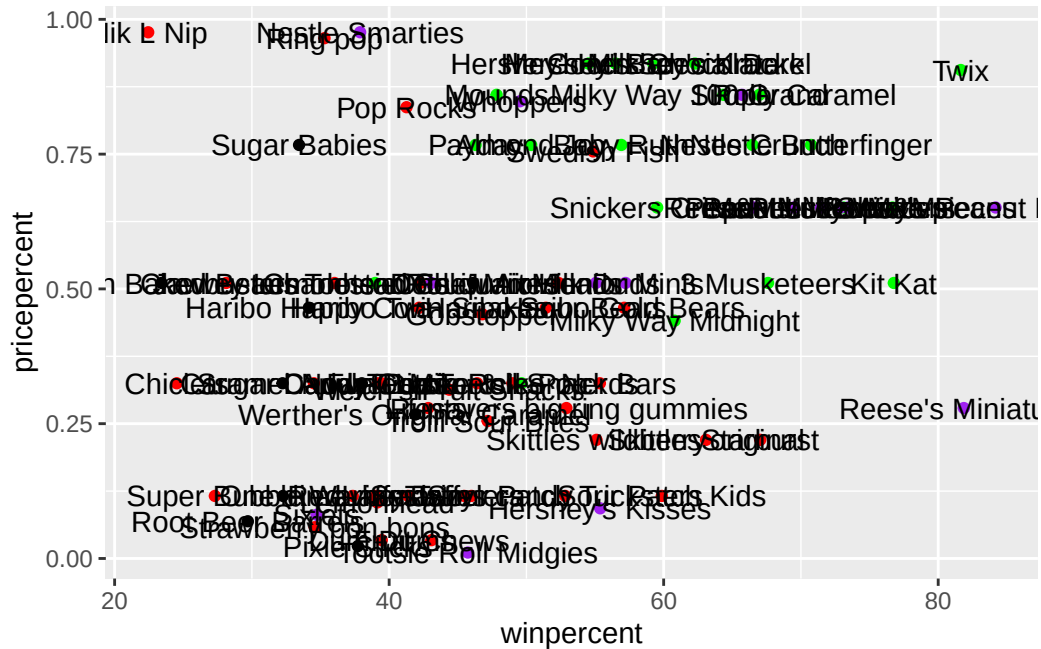
Q18. What is the best ranked fruity candy?

The best ranked fruity candy is star bursts

Taking a look at pricepercent

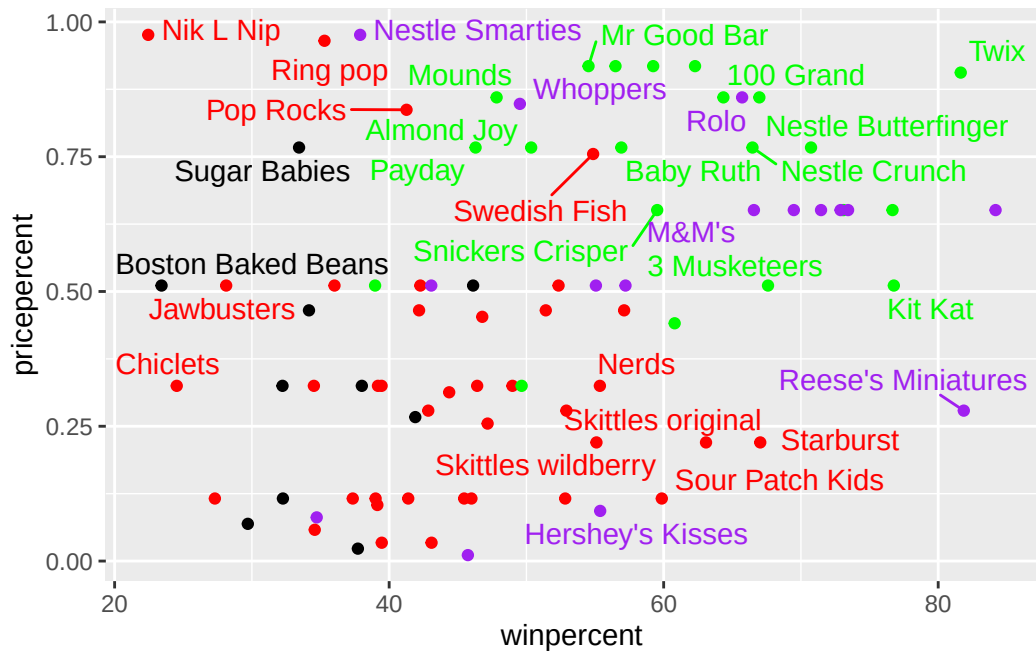
Now lets compare the win percent versus the price of the candy (its value)

```
ggplot(candy)+  
  aes(winpercent, pricepercent, label =rownames(candy))+  
  geom_point(color=my_cols)+  
  geom_text()
```



Lets fix the label text over plotting using the an add-on called **ggrepel** and the `geom_text_repel()` function.

```
library(ggrepel)
ggplot(candy)+
  aes(winpercent, pricepercent, label =rownames(candy))+
  geom_point(color=my_cols)+
  geom_text_repel(color=my_cols, max.overlaps=9)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

The best bang for your buck would be the reese's miniatures, you want to be low on the y-axis and high on the x axis.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
candy%>%
  arrange(-pricepercent)%>%
  select(pricepercent,winpercent)%>%
  head(n=5)
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534

Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

The top 5 most expensive Nik L Nip, Nestle smarties, Ring pop, Hershey's Krackel, Hershey Milk Chocolate. Of these 5 most expensive the most popular is Hershey's Krackel.

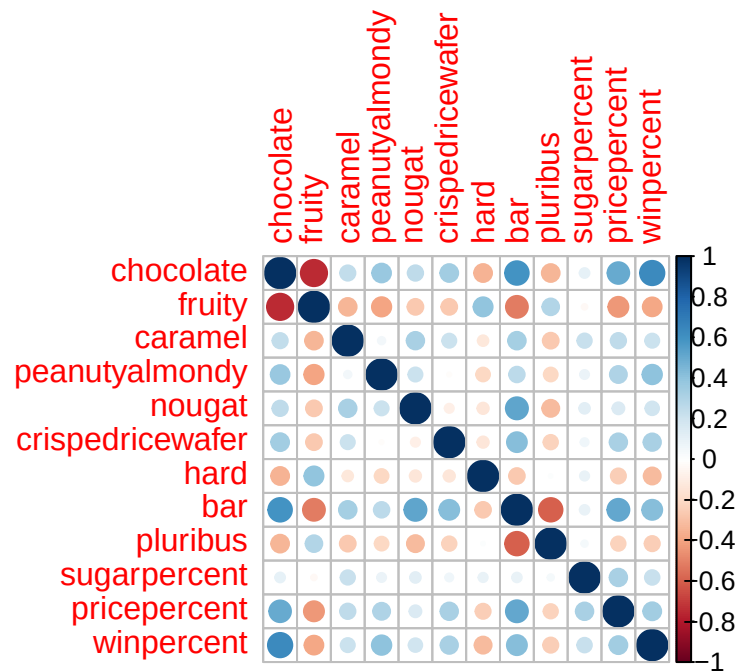
Exploring the correlation structure

We can calculate the pairwise correlation of all our columns.

```
cij<- cor(candy)
library(corrplot)
```

corrplot 0.95 loaded

```
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Very negative (strongly anti-correlated) is fruit and chocolate indicating they don't coexist (normally).

Q23. Use your corplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

Most likely PC1 will encompass whether the candy is chocolate or fruit based. This will have a lot of variance and split the data up nicely.

Principal Component Analysis (PCA)

We should set the `scale=TRUE` due to the difference between the scaling in the `winpercent` and the other variables. if `scale=FALSE` this would dominate the PCA results.

```
pca<- prcomp(candy,scale=TRUE)
summary(pca)
```

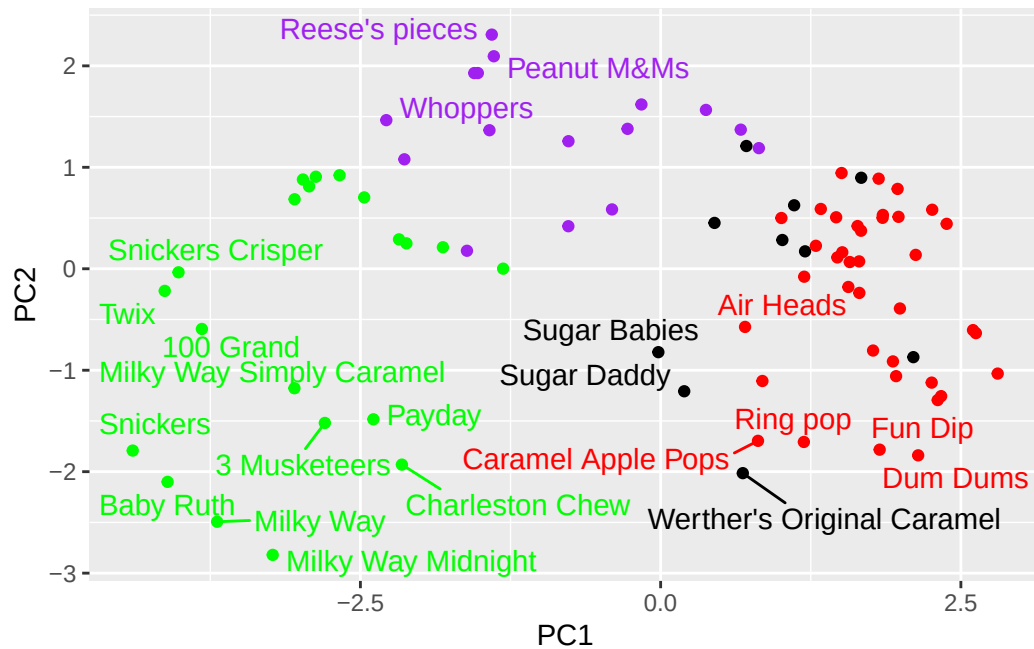
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

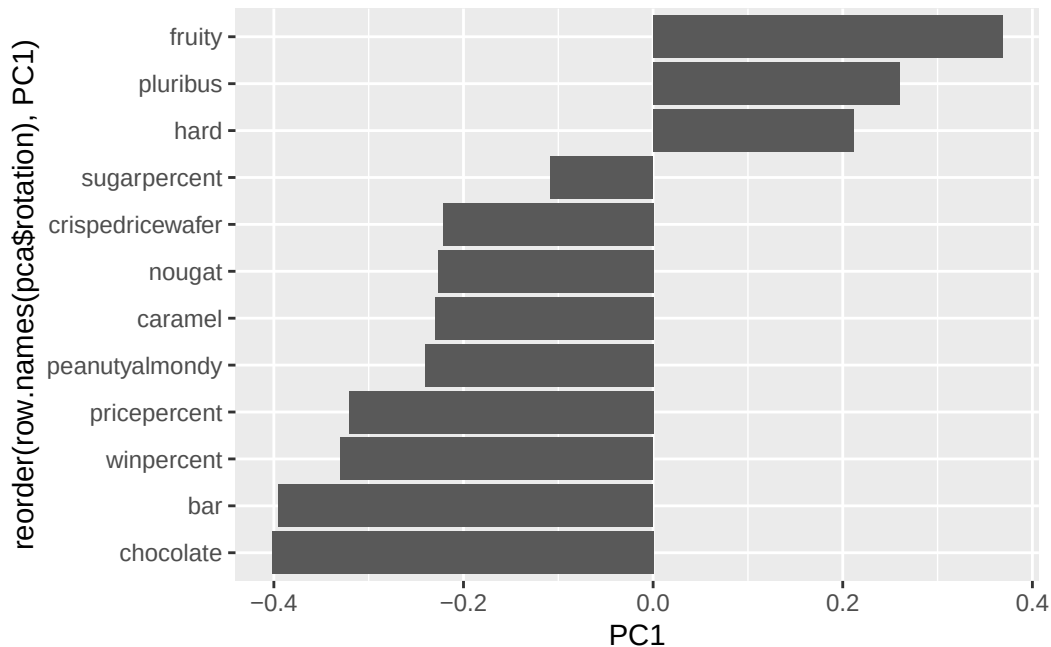
“Score plot” of PC1 versus PC2, this will show how the different candies are related to one another on the separate PC axis.

```
ggplot(pca$x)+
  aes(PC1, PC2, label=row.names(pca$x))+
  geom_point(color=my_cols)+
  geom_text_repel(color=my_cols, max.overlaps = 7)
```



The second major results figure from PCA. This is called “loadings plot” → how the original variables contribute to the PCA.

```
ggplot(pca$rotation)+
  aes(PC1,reorder(row.names(pca$rotation),PC1))+
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

Picked up strongly is pluribus, fruity and hard. These all are highlighted in the correlation plot. They showed negative correlation in a lot of the variables within the correlation plot.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

There is a high correlation between a candy being chocolate and being popular. The win percents tend to be the highest amongst groups with chocolates this can be seen in in the PCA of win percent and price percent, highlighted in green are chocolate candies with caramel, but these also tend to be the most expensive. If you are candy executive you would be able to charge the most and sell the most popular candy if it were chocolate.